

GK-Mapper: A Stability Framework for Gustafson-Kessel Fuzzy Mapper Graphs

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Abstract

Topological Data Analysis is the field that uses algebraic topology for data analysis, with the Mapper Algorithm that studies the structure of data after reducing the dimension of the dataset. There are several variants of Mapper, like Conventional Mapper, F-Mapper, and Shape Fuzzy-C Means Mapper. In this article, we extend the idea of the Shape Fuzzy-C Means Mapper graphs by introducing the Gustafson-Kessel Fuzzy Mapper Graphs algorithm, which replaces the spherical covers by ellipsoidal cover, which is useful for high dimensional datasets because real world datasets are not always symmetrical or spherical. We then develop the stability framework for the graphs produced by Gustafson Kessel Mapper graph and Shape Fuzzy C-Mean Mapper graph. We prove that the memberships depend smoothly on the fuzzifier, establish a precise condition for the existence of edges, and show that the graph is locally stable under small perturbations. We describe the critical event structure of graph changes in terms of threshold crossings of the membership functions and show that the graph is constant between consecutive critical events. When the threshold-crossing set is finite, this yields an eventual freezing threshold. Finally, we show empirically that the Gustafson Kessel Mapper is more stable than the Shape Fuzzy C Means for high-dimensional complex datasets.

Keywords: Gustafson Kessel Mapper, Gustafson Kessel FCM, Fuzzy Clustering, Mapper Algorithm, Topological Data Analysis, Stability Analysis, Fuzzifier Parameter, Simplicial Complex,

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1. Introduction

The Mapper algorithm [24] has become one of the tools in Topological Data Analysis for analysing the shape of complex, high-dimensional data. It transforms datasets into a graph, which is a 2D/3D representation of the datasets. It provides the summaries of connectivity, loops, and hidden geometric relationships that sometimes ML methods miss [9, 6]. It has been applied in various fields like bioinformatics, neuroscience, social networks, and many more. Some applications can be found in [16, 19, 20, 21, 22, 28]. Since it has so many applications, selecting its appropriate parameters is still a challenging task. Particularly, the choice of filter function, cover resolution, and overlap parameters highly influences the resulting graph. Several works have addressed this issue by studying stability conditions and robustness properties of Mapper graphs [1, 6, 7, 12].

The conventional Mapper [24] uses hard partitioning and rigid interval covers, but for real-world datasets, boundaries cannot always be rigid, and so Fuzzy variants of Mapper have been proposed. The F-Mapper algorithm [4] uses Fuzzy C -Mean (FCM) to generate overlapping covers. It creates soft cover rather than hard ones, which becomes more useful for real-world datasets. But it also contains the same number of parameters; the filter function, the fcm cover that requires the number of clusters and membership factor and the dbscan algorithm. To limit these parameters, The Shape Fuzzy C -Mean (SFCM) algorithm [5] combines FCM directly with the Mapper nerve construction. Thus, in this method, we only need the number of clusters and the threshold condition.

Although it solves a lot of our purpose, it again has two major limitations: The first one is that it uses FCM to generate cover, which assumes spherical cluster geometry. In many real world datasets, biological structures, and medical image clusters are non-spherical, and the Euclidean cover misrepresents the true cluster boundaries. Second, the stability of the SFCM graph with respect to the fuzzifier parameter m has not been studied. It is typically considered as $m = 2$ without theoretical justification [2], and without knowing whether small changes in m can affect the graph structure.

We address both limitations. We propose the *Gustafson-Kessel Fuzzy Mapper Graphs* (GK Mapper) algorithm, which replaces the Euclidean FCM cover of SFCM with a cover generated by the Gustafson-Kessel FCM (GK-

FCM) algorithm [15, 2], which considers an ellipsoidal structure for the cluster. We then develop a stability framework for GK Mapper and empirically show that GK-Mapper often performs well across several aspects for which theoretical foundations are developed in this paper. The contributions of the paper are as follows:

1. We propose the GK-Mapper algorithm, which modifies the cover construction of the SFCM algorithm with the Gustafson-Kessel-based cover.
2. We characterise the edgeless-zone boundary by the critical threshold $t_{\text{crit}}(m) = \max_i \max_{j \neq k} \min\{u_{ij}(m), u_{ik}(m)\}$, above which the graph becomes edgeless.
3. We then prove a local structural stability theorem with a computable stability radius r^* . This radius indicates how far one can vary a chosen value of m , obtaining the same Mapper graph.
4. We show that the GK-Mapper graph can change only at threshold-crossing events and is constant between consecutive critical events. We further provide a crossing-count bound for the number of critical events and recover the estimate $|\mathcal{T}| \leq nc$ under a single-crossing condition.
5. We empirically show that GK-Mapper performs well in all these cases as compared to SFCM.

The rest of this paper is organised as follows. Section 2 introduces the necessary background and definitions. Section 3 proposes the GK-Mapper algorithm. Section 4 establishes membership regularity for both FCM and GK-FCM. Section 5 presents the main stability framework. Section 6 provides empirical validation on synthetic and real-world datasets. Section 7 discusses the implications and limitations of the framework. Section 8 concludes the paper.

2. Background

Fuzzy set theory [29] extends classical set membership by allowing each element to belong to a set with a degree in the interval $[0, 1]$ rather than in a strictly binary manner. It is useful for real-world data, where cluster boundaries are often vague, overlapping, or uncertain. In clustering, such partial

memberships provide a natural way to model ambiguity at the interfaces between groups.

The Fuzzy C -Mean (FCM) algorithm [3] is one of the most widely used fuzzy clustering methods. It assigns memberships to all clusters and determines cluster centres by minimising a weighted objective function. This soft partitioning makes FCM more flexible than hard partitioning clustering methods, especially when the data contains overlap or gradual transitions between groups. In the standard formulation, however, FCM relies on the Euclidean distance, which implicitly favours approximately spherical clusters.

This section provides a brief overview of the definitions and notations related to the GK-FCM, Mapper, and SFCM algorithms [2, 15, 24, 5].

Definition 1 (Gustafson-Kessel FCM [15, 2]). The Gustafson-Kessel FCM (GK-FCM) algorithm addresses the spherical-cluster limitation of the standard FCM by replacing the Euclidean metric with a cluster-adaptive Mahalanobis-type distance [11]. Each cluster is allowed to adapt its shape according to the local covariance structure of the data, making GK-FCM particularly suitable for datasets with ellipsoidal or directionally stretched clusters. This adaptive geometry provides the foundation for the cover used in GK-Mapper. Further detail on clustering variants and their applications can be found in [2, 17, 26, 27].

Definition 2 (Mapper Algorithm [24]). Let X be a dataset, let $f : X \rightarrow \mathbb{R}^d$ be a continuous filter function, and let $\mathcal{U} = \{U_a\}$ be an open cover of $f(X)$. For each $U_a \in \mathcal{U}$, apply a clustering algorithm to the preimage $f^{-1}(U_a)$, producing clusters $\{C_{a,1}, C_{a,2}, \dots\}$. The *Mapper complex* is the simplicial complex where each cluster $C_{a,i}$ is a node and two nodes are connected by an edge whenever $C_{a,i} \cap C_{b,j} \neq \emptyset$. More generally, a k -simplex is added whenever $k+1$ clusters have a common nonempty intersection.

Definition 3 (F-Mapper Algorithm [4]). Let X be a finite dataset in a metric space, and let $f : X \rightarrow \mathbb{R}$ be a continuous filter. F-Mapper partitions $f(X)$ into N fuzzy clusters by FCM, producing membership degrees $u_{ij} \in [0, 1]$ satisfying $\sum_{j=1}^N u_{ij} = 1$. For a threshold $t \in [0, 1]$, define fuzzy cover intervals $U_j = \{f(x_i) : u_{ij} \geq t\}$ and pullback sets $f^{-1}(U_j)$. Each pullback set is clustered into connected components, and the *F-Mapper complex* is the nerve of these components.

Definition 4 (SFCM Algorithm [5]). Let $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^p$, $c \geq 2$, $m > 1$, and $t \in (T_0, T_1]$, where $T_0 = \min_{i,j} u_{ij}$ and $T_1 = \min_i \max_j u_{ij}$. SFCM minimises the FCM objective $J_m(U, V) = \sum_{i=1}^n \sum_{j=1}^c (u_{ij})^m \|x_i - v_j\|^2$, producing clusters $C_j(t) = \{x_i : u_{ij}(m) \geq t\}$. The *SFCM graph* is $G_t(m) = (V, E(m))$ where $V = \{1, \dots, c\}$ and $E(m) = \{(j, k) : j \neq k, C_j \cap C_k \neq \emptyset\}$. The complete algorithm is given in Algorithm 2.

3. GK-Fuzzy Mapper Algorithm

The SFCM algorithm uses the Euclidean-distance assumption from FCM, which naturally favours spherical cluster shapes. However, many real-world datasets contain clusters that are elongated, ellipsoidal, or otherwise non-spherical. To address this limitation, we introduce the Gustafson Kessel Mapper (GK-Mapper) algorithm. This method removes the spherical constraint by replacing the Euclidean cover used in SFCM with a geometry-adaptive cover derived from the Gustafson-Kessel FCM algorithm [15]. Compared with SFCM, GK-Mapper uses the same number of parameters. The only change is that Euclidean distance is replaced by a cluster-adaptive distance [2, Theorem 22.1].

The detailed computational procedure is presented in Algorithm 3. The algorithm follows the standard fuzzy clustering framework, where cluster centres and memberships are iteratively updated using the Gustafson-Kessel adaptive distance. After convergence, the fuzzy memberships are thresholded to construct the adaptive cover, and the Mapper graph is obtained by connecting clusters with nonempty intersections.

4. Membership Regularity

Before developing the stability framework, we establish that the GK-FCM membership function depends smoothly on the fuzzifier parameter m along the optimisation path. This regularity is the foundation on which all subsequent results rest. The following proposition is stated for the moving-centre setting, where the centres $v_j(m)$ and (in the GK-Mapper case) the adaptive matrices $A_j(m)$ depend on m . The arguments depend only on the composition structure of the membership formula, and not on the specific distance used. Before stating the main regularity result, we fix the standing assumptions that govern Sections 4 and 5.

Assumption 1 (H1-Continuity). *For each $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, c\}$, the membership function $m \mapsto u_{ij}(m)$ is continuous at m_0 .*

Assumption 2 (H2- C^1 Optimisation Path). *The optimisation path $m \mapsto V(m) = \{v_1(m), \dots, v_c(m)\}$ is C^1 on an open interval $\mathcal{I} \ni m_0$. In the GK-Mapper case, the adaptive matrices $m \mapsto A_j(m)$ are additionally C^1 on $\mathcal{I}$ for every $j \in \{1, \dots, c\}$. Furthermore, the non-degeneracy condition*

$$d_{il}(m_0) > 0 \quad \forall i \in \{1, \dots, n\}, l \in \{1, \dots, c\} \quad (1)$$

holds, that is, no data point coincides with any cluster centre at m_0 .

Proposition 1 (Membership Regularity Along the Optimisation Path). *Let $m \mapsto V(m) = \{v_1(m), \dots, v_c(m)\}$ be a C^1 path of cluster centres on an interval $\mathcal{I} \subset (1, \infty)$. In the GK-Mapper case, also assume that $m \mapsto A_j(m)$ is C^1 for every $j \in \{1, \dots, c\}$. Define*

$$d_{ij}(m) = \begin{cases} \|x_i - v_j(m)\|, & \text{SFCM,} \\ \sqrt{(x_i - v_j(m))^\top A_j(m) (x_i - v_j(m))}, & \text{GK,} \end{cases}$$

and assume $d_{ij}(m) > 0$ for all i, j and all $m \in \mathcal{I}$. Set

$$r_{ijk}(m) = \frac{d_{ij}(m)}{d_{ik}(m)}, \quad b(m) = \frac{2}{m-1},$$

and

$$D_{ij}(m) = \sum_{k=1}^c r_{ijk}(m)^{b(m)}.$$

Then

$$u_{ij}(m) = \frac{1}{D_{ij}(m)}$$

satisfies:

- (i) $u_{ij}(m) \in (0, 1)$ for all $m \in \mathcal{I}$;
- (ii) $u_{ij}(m)$ is of class C^1 on $\mathcal{I}$;
- (iii) $\sum_{j=1}^c u_{ij}(m) = 1$ for all i and m ;

(iv) $u_{ij}(m)$ is differentiable with

$$u'_{ij}(m) = T_1(m) + T_2(m),$$

where

$$T_1(m) = \frac{2u_{ij}(m)^2}{(m-1)^2} \sum_{k=1}^c r_{ijk}(m)^{b(m)} \ln r_{ijk}(m),$$

and $T_2(m)$ contains the contribution coming from the motion of the centres and, in the GK-Mapper case, the adaptive matrices.

More explicitly, for SFCM,

$$T_2^{\text{SFCM}}(m) = \frac{2u_{ij}(m)^2}{m-1} \sum_{l=1}^c \frac{w_{ijl}}{d_{il}(m)^2} (x_i - v_l)^\top \dot{v}_l,$$

whereas for GK-Mapper,

$$\begin{aligned} T_2^{\text{GK}}(m) &= \frac{2u_{ij}(m)^2}{m-1} \sum_{l=1}^c \frac{w_{ijl}}{d_{il}(m)^2} \\ &\times \left[(x_i - v_l)^\top A_l \dot{v}_l - \frac{1}{2} (x_i - v_l)^\top \dot{A}_l (x_i - v_l) \right], \end{aligned}$$

with

$$w_{ijl} = \begin{cases} D_{ij}(m) - 1, & l = j, \\ -r_{ijl}(m)^{b(m)}, & l \neq j. \end{cases}$$

Proof. Since $d_{ij}(m) > 0$ and the distance functions are C^1 in m under the stated assumptions, each ratio $r_{ijk}(m)$ is strictly positive and C^1 . For $m > 1$, $b(m) > 0$, so each summand $r_{ijk}(m)^{b(m)}$ is positive and $D_{ij}(m) > 0$. Since the term $k = j$ equals 1 and $c \geq 2$, we have $D_{ij}(m) > 1$. Therefore $u_{ij}(m) \in (0, 1)$.

Next, each summand can be written as

$$r_{ijk}(m)^{b(m)} = \exp(b(m) \ln r_{ijk}(m)).$$

Since both $b(m)$ and $\ln r_{ijk}(m)$ are C^1 on $\mathcal{I}$, it follows that D_{ij} is C^1 . Hence $u_{ij} = D_{ij}^{-1}$ is also C^1 on $\mathcal{I}$.

The identity $\sum_{j=1}^c u_{ij}(m) = 1$ is the standard normalisation property of fuzzy memberships [2, Def. 5.1].

Finally, differentiating $u_{ij} = D_{ij}^{-1}$ gives

$$u'_{ij}(m) = -u_{ij}(m)^2 D'_{ij}(m).$$

Using $b'(m) = -2/(m-1)^2$ and the chain rule,

$$\begin{aligned} \frac{d}{dm} [r_{ijk}(m)^{b(m)}] &= r_{ijk}(m)^{b(m)} \left[-\frac{2}{(m-1)^2} \ln r_{ijk}(m) \right. \\ &\quad \left. + b(m) \frac{\dot{r}_{ijk}(m)}{r_{ijk}(m)} \right]. \end{aligned}$$

Therefore,

$$\begin{aligned} D'_{ij}(m) &= -\frac{2}{(m-1)^2} \sum_{k=1}^c r_{ijk}(m)^{b(m)} \ln r_{ijk}(m) \\ &\quad + b(m) \sum_{k=1}^c r_{ijk}(m)^{b(m)} \frac{\dot{r}_{ijk}(m)}{r_{ijk}(m)}. \end{aligned}$$

Substituting this expression into $u'_{ij} = -u_{ij}^2 D'_{ij}$ gives the decomposition $u'_{ij}(m) = T_1(m) + T_2(m)$. The stated forms of T_2^{SFCM} and T_2^{GK} follow by differentiating the corresponding distance functions and collecting the terms associated with each moving centre and adaptive matrix. $\square$

Corollary 1 (Non-monotonicity Along the Optimisation Path). *Under the hypotheses of Proposition 1, each membership function $u_{ij}(m)$ is C^1 on $\mathcal{I}$. However, along the optimisation path, $u_{ij}(m)$ is not necessarily monotone.*

Indeed, the derivative has the form

$$u'_{ij}(m) = T_1(m) + T_2(m),$$

where T_1 is determined by the fuzzifier-dependent exponent and distance ratios, while T_2 contains the effect of centre motion and, in GK-Mapper, adaptive-matrix motion. Since T_2 may have either sign, the sign of $u'_{ij}(m)$ is not determined by the distance ratios alone. Consequently, monotonicity of $u_{ij}(m)$ cannot be assumed without additional restrictions on the optimisation path.

Proof. The result follows directly from the derivative decomposition in Proposition 1. The term T_2 depends on $\dot{v}_l$ and, in the GK-Mapper case, on $\dot{A}_l$.

These quantities may vary in direction and magnitude along the optimisation path. Hence T_2 may be positive, negative, or zero, and no general sign condition for $u'_{ij}(m)$ follows from the membership formula alone. Therefore $u_{ij}(m)$ need not be monotone on $\mathcal{I}$. $\square$

Remark 1. Proposition 1 and Corollary 1 apply to both SFCM and GK-Mapper. The only difference is the form of the distance function: SFCM uses Euclidean distances, while GK-Mapper uses cluster-adaptive Gustafson-Kessel distances. Hence, the subsequent stability results apply to both constructions, with the GK-Mapper case including the additional contribution from the evolution of the adaptive matrices.

5. Main Results

In this section, we present the main theoretical results that describe how the GK-Mapper graphs change as the fuzzifier m varies. These results identify the parameter setting in which the graph carries structural information, establishes a local stability zone around any reference value of m , quantify graph variation under small perturbations, and describe the critical event structure of the graph along the moving centre optimisation path.

We begin by characterising the Edgeless Zone (Section 5.1), where the graph has no edges. We then establish a Stability Zone (Section 5.2) in which the graph remains unchanged under small changes in m . We next analyse the Instability Zone (Section 5.3) and derive an upper bound on edge changes. Finally, we describe the Critical Event Structure (Section 5.4) and analyse when the graph eventually freezes beyond a finite threshold m^{**} .

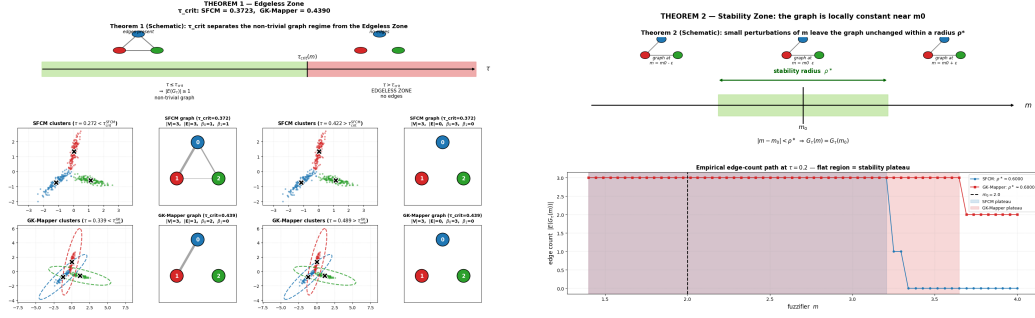
5.1. The Edgeless Zone

A parameter pair (m, t) lies in the *Edgeless Zone* if $G_t(m)$ has zero edges, then the graph carries no structural information. Theorem 1 establishes the necessary and sufficient condition for avoiding the Edgeless Zone. A visual illustration is provided in Fig. 1a.

Theorem 1 (Edgeless Zone). *Let $G_t(m)$ be the GK-Mapper graph. Then $G_t(m)$ has at least one edge if and only if*

$$t \leq \max_i \max_{j \neq k} \min\{u_{ij}(m), u_{ik}(m)\}. \quad (2)$$

Equivalently, $G_t(m)$ has no edges iff $t > \max_i \max_{j \neq k} \min\{u_{ij}(m), u_{ik}(m)\}$.



(a) Visual illustration of Theorem 1 for the anisotropic ellipsoidal dataset at $m_0 = 2.0$. The critical threshold $t_{crit}(m)$ separates the non-trivial graph regime from the Edgeless Zone.

(b) Visual illustration of Theorem 2 for the anisotropic ellipsoidal dataset at $m_0 = 2.0$ and $t = 0.2$. The stability radius r^* defines a local interval where no membership value crosses t , so $G_t(m) = G_t(m_0)$. The flat edge-count paths confirm the predicted local graph constancy.

Figure 1: Empirical illustrations of Theorems 1 and 2. Subfigure 1a shows the Edgeless Zone transition, while Subfigure 1b shows local graph stability around $m_0 = 2.0$.

Proof. ($\Rightarrow$) Suppose $G_t(m)$ contains an edge; then $C_j \cap C_k \neq \emptyset$ for some $j \neq k$. Let x_i be a point in this intersection. Then $u_{ij}(m) \geq t$ and $u_{ik}(m) \geq t$, so $\min\{u_{ij}(m), u_{ik}(m)\} \geq t$, giving (2).

($\Leftarrow$) If (2) holds, there exist i, j, k with $j \neq k$ such that $u_{ij}(m) \geq t$ and $u_{ik}(m) \geq t$. Therefore $x_i \in C_j \cap C_k$ and an edge (j, k) exists. $\square$

We define $t_{crit} = \max_i \max_{j \neq k} \min\{u_{ij}(m), u_{ik}(m)\}$ as the critical threshold above which the graph enters the Edgeless Zone.

5.2. The Stability Zone

After identifying the region where the graph becomes edgeless, we focus on the area where the graph structure is preserved under small perturbations of the fuzzifier m . Intuitively, if the membership values do not cross the threshold t , the induced cover and hence the graph topology remain unchanged. The following theorem gives a radius r^* for which the graph remains unchanged for a chosen m . Its behaviour described is illustrated in Fig. 1b.

Throughout Theorem 2, we assume

$$d_{il}(m_0) > 0, \quad \forall i \in \{1, \dots, n\}, l \in \{1, \dots, c\}, \quad (3)$$

that is, no data point coincides with any cluster centre at m_0 . This ensures that every distance denominator appearing in the proof is strictly positive. This is the standard non-degeneracy condition in FCM [2].

Moreover, for each $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, c\}$, let $d_{ij}(m)$ denote the distance from x_i to the cluster centre $v_j(m)$ under the relevant metric:

$$d_{ij}(m) = \begin{cases} \|x_i - v_j(m)\|, & \text{SFCM,} \\ \sqrt{(x_i - v_j(m))^\top A_j(m) (x_i - v_j(m))}, & \text{GK-Mapper.} \end{cases} \quad (4)$$

Define

$$r_{ijk}(m) = \frac{d_{ij}(m)}{d_{ik}(m)}, \quad D_{ij}(m) = \sum_{k=1}^c r_{ijk}(m)^{b(m)},$$

$$b(m) = \frac{2}{m-1}, \quad u_{ij}(m) = \frac{1}{D_{ij}(m)}.$$

Note that $r_{ijj}(m) \equiv 1$, and therefore

$$D_{ij}(m) = 1 + \sum_{k \neq j} r_{ijk}(m)^{b(m)}.$$

Theorem 2 (Local Stability Zone). *Let $m_0 > 1$ and $t \in (0, 1)$ satisfy*

$$u_{ij}(m_0) \neq t \quad \text{for all } i \in \{1, \dots, n\}, j \in \{1, \dots, c\}.$$

Assume the non-degeneracy condition (3) and suppose that the hypotheses of Proposition 1 hold on a neighbourhood of m_0 . Then there exists $r^ > 0$ such that*

$$G_t(m) = G_t(m_0) \quad \text{whenever } |m - m_0| < r^*. \quad (5)$$

Hence, the SFCM and GK-Mapper graphs are locally constant with respect to the fuzzifier parameter near every non-threshold value m_0 .

Proof. By Proposition 1, each membership function $u_{ij}(m)$ is continuous, indeed C^1 , in a neighbourhood of m_0 . Since $u_{ij}(m_0) \neq t$, define the positive threshold margin

$$d_{ij}^{\text{gap}} := |u_{ij}(m_0) - t| > 0.$$

By continuity of u_{ij} at m_0 , there exists $r_{ij} > 0$ such that

$$|u_{ij}(m) - u_{ij}(m_0)| < d_{ij}^{\text{gap}} \quad \text{whenever } |m - m_0| < r_{ij}.$$

Since there are only finitely many pairs (i, j) , define

$$r^* := \min_{1 \leq i \leq n, 1 \leq j \leq c} r_{ij}.$$

Then $r^* > 0$. Hence, for every m satisfying $|m - m_0| < r^*$, we have

$$|u_{ij}(m) - u_{ij}(m_0)| < |u_{ij}(m_0) - t| \quad \text{for all } i, j.$$

Therefore $u_{ij}(m)$ and $u_{ij}(m_0)$ lie on the same side of the threshold t . Consequently,

$$\mathbf{1}[u_{ij}(m) \geq t] = \mathbf{1}[u_{ij}(m_0) \geq t] \quad \text{for all } i, j.$$

Thus the thresholded cluster sets $C_j(t, m) = \{x_i : u_{ij}(m) \geq t\}$ remain unchanged for all j . Since the edge set of $G_t(m)$ is determined by the nonempty intersections $C_a(t, m) \cap C_b(t, m) \neq \emptyset$, the edge set also remains unchanged. Therefore $G_t(m) = G_t(m_0)$ whenever $|m - m_0| < r^*$. $\square$

Remark 2. The proof above establishes the existence of a local stability radius. A conservative computable estimate can be obtained from the derivative formula in Proposition 1. Let $J = [m_0 - \text{tol}, m_0 + \text{tol}]$ be a compact neighbourhood contained in the interval of regularity, and set $M_{ij}(\text{tol}) := \sup_{s \in J} |u'_{ij}(s)|$. By Proposition 1, u'_{ij} is continuous, so $M_{ij}(\text{tol}) < \infty$. The mean value theorem gives $|u_{ij}(m) - u_{ij}(m_0)| \leq M_{ij}(\text{tol})|m - m_0|$. Hence, one may take

$$r^* = \min_{i,j} \left\{ \text{tol}, \frac{|u_{ij}(m_0) - t|}{M_{ij}(\text{tol})} \right\},$$

with the convention that if $M_{ij}(\text{tol}) = 0$, the corresponding term is taken as tol .

Remark 3. Theorem 2 establishes that both GK-Mapper and SFCM graphs are locally stable near any non-threshold fuzzifier value m_0 , with computable stability radii $r^{*,\text{GK}}$ and $r^{*,\text{SFCM}}$. Whether GK-Mapper or SFCM achieves a larger stability radius depends on the underlying cluster geometry through the distance ratios, the rates of evolution of the shape matrices and many other factors. For complex datasets, the empirical evidence in Section 6 suggests that GK-Mapper can produce larger stability regions than SFCM.

5.3. The Instability Zone

Having established local stability, we now quantify how many edges can change when the fuzzifier is perturbed from m to $m+h$. For each membership entry, define the threshold indicator

$$\text{ind}_{ij}(m) = \mathbf{1}[u_{ij}(m) \geq t]. \quad (6)$$

For a pair of clusters (a, b) , define the witness count

$$I_{ab}(m) = \sum_{i=1}^n \text{ind}_{ia}(m) \text{ind}_{ib}(m). \quad (7)$$

Thus, $I_{ab}(m)$ counts the number of data points simultaneously belonging to the thresholded clusters a and b . Hence, the edge (a, b) exists in $G_t(m)$ if and only if $I_{ab}(m) > 0$. This behaviour is illustrated in Fig. 2a.

Lemma 1. *An edge (a, b) changes between $G_t(m)$ and $G_t(m+h)$ if and only if*

$$I_{ab}(m) I_{ab}(m+h) = 0 \quad \text{and} \quad I_{ab}(m) + I_{ab}(m+h) > 0.$$

Consequently, $G_t(m) \neq G_t(m+h)$ if and only if the above condition holds for at least one pair (a, b) .

Proof. The edge (a, b) exists exactly when $I_{ab}(m) > 0$. Therefore, the edge changes between m and $m+h$ precisely when one of the two witness counts $I_{ab}(m)$ and $I_{ab}(m+h)$ is positive and the other is zero, which is equivalent to the stated conditions. The graph changes if and only if at least one edge changes. $\square$

Theorem 3 (Edge-Change Bound). *Define the threshold-crossing set*

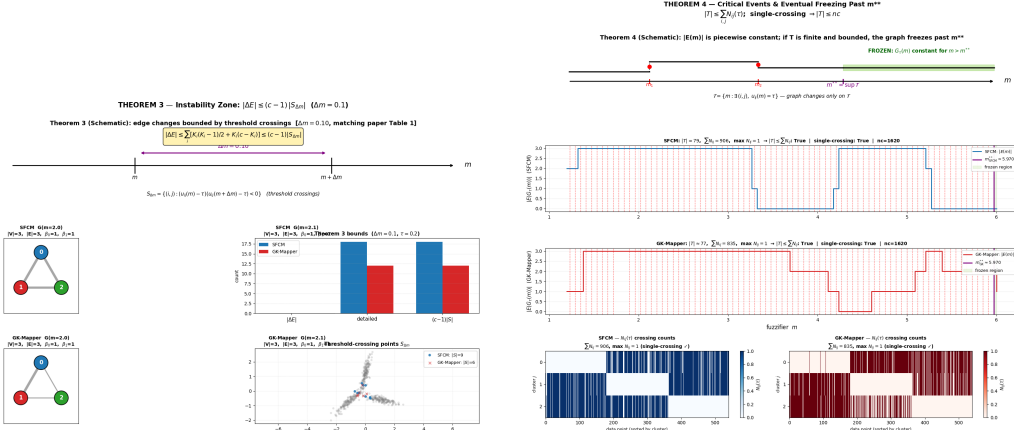
$$S_h = \{(i, j) : (u_{ij}(m) - t)(u_{ij}(m+h) - t) < 0\}.$$

For each i , let $K_i = |\{j : (i, j) \in S_h\}|$. Then the number of edge changes satisfies

$$|E_{\text{chg}}| \leq \sum_{i=1}^n \left[\binom{K_i}{2} + K_i(c - K_i) \right] \leq (c-1)|S_h|. \quad (8)$$

Proof. An edge (a, b) can change only if, for some data point x_i , at least one of the indicators ind_{ia} or ind_{ib} changes between m and $m+h$. Such a change can occur only when $(i, a) \in S_h$ or $(i, b) \in S_h$.

Fix a data point x_i and suppose that K_i of its membership entries cross the threshold. The affected cluster pairs are of two types. First, both indices may belong to the crossing set, giving at most $\binom{K_i}{2}$ pairs. Second, exactly one index may belong to the crossing set, giving at most $K_i(c - K_i)$ pairs. Hence the number of edge pairs affected by x_i is at most $\binom{K_i}{2} + K_i(c - K_i)$. Summing over all data points gives the first inequality. Since $\binom{K_i}{2} + K_i(c - K_i) \leq K_i(c - 1)$, summing yields $(c - 1) \sum_i K_i = (c - 1)|S_h|$. $\square$



(a) Visual illustration of Theorem 3 for the anisotropic ellipsoidal dataset. Edge changes in the t -superlevel cluster co-occurrence graph are controlled by membership threshold crossings between $m = 2.0$ and $m = 2.2$. The SFCM and GK-Mapper graphs remain unchanged, confirming that the theorem gives a conservative upper bound on graph instability.

(b) Visual illustration of Theorem 4 for the t -superlevel cluster co-occurrence graph. Graph changes occur only at critical fuzzifier values satisfying $u_{ij}(m) = t$. The edge-count paths are piecewise constant, with the frozen regime appearing after m^* .

Figure 2: Empirical illustrations of Theorems 3 and 4. Subfigure 2a shows the conservative instability bound through membership threshold crossings, while Subfigure 2b shows critical-event structure and eventual graph freezing.

5.4. Critical Events and Eventual Freezing

Having shown that edge changes are controlled by membership threshold crossings, we now describe the critical-event structure of the graph as the fuzzifier m varies. Since the memberships are evaluated along the optimisation path, they need not be monotone in m . Therefore, a membership value may cross the threshold t more than once. We formulate the result in terms of the actual threshold-crossing events.

Let $\mathcal{I} \subset (1, \infty)$ be the interval of fuzzifier values under consideration. Define the critical-event set

$$\mathcal{T} = \{m \in \mathcal{I} : \exists (i, j) \text{ such that } u_{ij}(m) = t\}.$$

For each pair (i, j) , define the threshold-crossing count

$$N_{ij}(t) = \# \{m \in \mathcal{I} : u_{ij}(m) = t\}.$$

Thus, $N_{ij}(t)$ records the number of times the membership of x_i in cluster j reaches the threshold t on $\mathcal{I}$.

Theorem 4 (Critical Events and Eventual Freezing). *Assume that each membership function $u_{ij}(m)$ is continuous on $\mathcal{I}$. Then the following statements hold.*

- (i) $G_t(m)$ can change only at values $m \in \mathcal{T}$. Equivalently, $G_t(m)$ is constant on every connected component of $\mathcal{I} \setminus \mathcal{T}$.
- (ii) If $N_{ij}(t) < \infty$ for every pair (i, j) , then

$$|\mathcal{T}| \leq \sum_{i=1}^n \sum_{j=1}^c N_{ij}(t).$$

- (iii) If $N_{ij}(t) \leq 1$ for all (i, j) , then $|\mathcal{T}| \leq nc$.
- (iv) If $\mathcal{T}$ is finite and bounded above in $\mathcal{I}$, and if $m^{**} := \sup \mathcal{T}$, then $G_t(m)$ is constant on $\mathcal{I} \cap (m^{**}, \infty)$.

Proof. For each i, j , define the threshold indicator $\text{ind}_{ij}(m) = \mathbf{1}[u_{ij}(m) \geq t]$, so that $x_i \in C_j(t, m) \iff \text{ind}_{ij}(m) = 1$. An edge (a, b) exists in $G_t(m)$ if and only if there exists some x_i such that $\text{ind}_{ia}(m) \text{ind}_{ib}(m) = 1$.

(i) Suppose m_1 and m_2 lie in the same connected component of $\mathcal{I} \setminus \mathcal{T}$. Then $u_{ij}(m) \neq t$ throughout the interval between m_1 and m_2 . Since u_{ij} is continuous, it cannot move from one side of t to the other without attaining the value t . Hence $\text{ind}_{ij}(m_1) = \text{ind}_{ij}(m_2)$ for all i, j , so the edge set of $G_t(m)$ is unchanged. Consequently, graph changes can occur only at values in $\mathcal{T}$.

(ii) For each fixed pair (i, j) , the equation $u_{ij}(m) = t$ has exactly $N_{ij}(t)$ solutions on $\mathcal{I}$. Hence $|\mathcal{T}| \leq \sum_{i=1}^n \sum_{j=1}^c N_{ij}(t)$, with inequality because several memberships may cross t at the same value of m .

(iii) If $N_{ij}(t) \leq 1$ for all (i, j) , then $\sum_{i,j} N_{ij}(t) \leq nc$, giving $|\mathcal{T}| \leq nc$.

(iv) Since $\mathcal{T}$ is finite, no critical event occurs in $\mathcal{I} \cap (m^{**}, \infty)$. By part (i), the graph is constant on every connected component of $\mathcal{I} \setminus \mathcal{T}$. Therefore $G_t(m)$ is constant on $\mathcal{I} \cap (m^{**}, \infty)$. $\square$

Remark 4. Theorem 4 does not assume that the memberships are monotone in m . It only requires continuity. The bound in part (ii) uses the actual number of threshold-crossing events, while part (iii) gives the simpler estimate $|\mathcal{T}| \leq nc$ only when each membership reaches the threshold at most once on $\mathcal{I}$. Thus, the eventual freezing point m^{**} is conditional on the critical-event set being finite and bounded above.

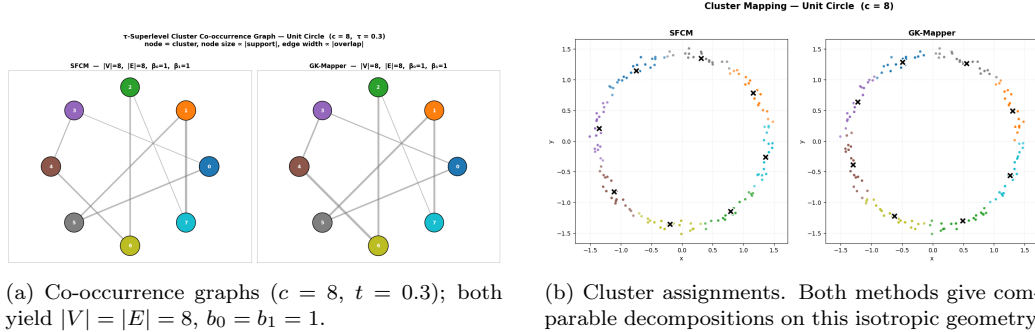


Figure 3: Unit circle dataset: SFCM (left) and GK-Mapper (right).

6. Empirical Validation

We evaluate GK-Mapper against SFCM on five datasets—Circle, Anisotropic Ellipsoidal, Stanford Bunny, UCI Handwritten Digits, and Wisconsin Breast Cancer to test the stability framework of Section 5. The reported quantities are the critical threshold t_{crit} (Theorem 1; largest t admitting at least one edge), the empirical stability radius r^* (Theorem 2; local robustness in m), the edge instability $|E_{\text{chg}}|$ under perturbation h , the Theorem 3 detailed/simple bounds, and standard clustering metrics (Silhouette, ARI, matching score). All reference fuzzifiers m_0 are selected automatically from the search grid. Aggregate results appear in Table 1.

6.1. Circle Dataset

A synthetic benchmark of 150 points on the unit circle in $\mathbb{R}^2$ with Gaussian noise ($s = 0.05$) and eight angular sectors as ground-truth labels. Both methods use $c = 8, t = 0.30$, with $m_0 = 4.457$ (SFCM) and $m_0 = 2.286$ (GK-Mapper). GK-Mapper raises t_{crit} from 0.4016 to 0.4652 and increases r^* from 0.0229 to 0.0833. Under $h = 0.10$, both methods yield $|E_{\text{chg}}| = 0$, indicating that the graph structure is unchanged under the selected perturbation.

Clustering metrics show a mixed but favourable trend for GK-Mapper. SFCM has a slightly higher Silhouette score (0.495 vs. 0.480), whereas GK-Mapper gives a substantially higher ARI (0.814 vs. 0.513) and matching score (0.913 vs. 0.713). Both methods recover the expected circular topology, with $|V| = 8, |E| = 8$, and $b_0 = b_1 = 1$; see Fig. 3. Thus, on this dataset, GK-Mapper preserves the same topological structure as SFCM while improving the empirical stability radius and external label agreement.

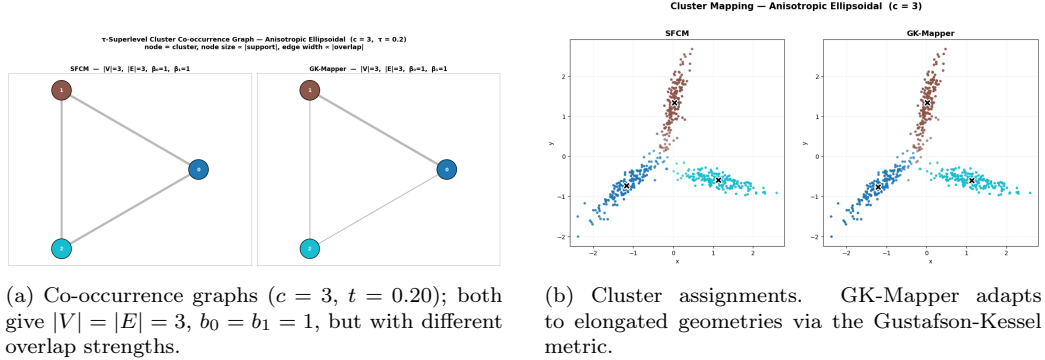


Figure 4: Anisotropic ellipsoidal dataset.

6.2. Anisotropic Ellipsoidal Dataset

Three Gaussian clusters (180 points each) in $\mathbb{R}^2$ with aspect ratios 5:1, 8:1, 4:1, deliberately violating spherical-cluster assumptions [23]. Both methods use $c = 3$, $t = 0.20$, and $m_0 = 1.200$. GK-Mapper improves t_{crit} from 0.3723 to 0.4390 and increases r^* from 0.0132 to 0.0333. Under the selected perturbation $h = 0.10$, both methods give $|E_{\text{chg}}| = 0$, so no edge change is observed.

Both methods produce the same graph-level topological summary: $|V| = 3$, $|E| = 3$, $b_0 = 1$, and $b_1 = 1$ (Fig. 4). The clustering metrics are also very close. SFCM has a slightly higher Silhouette score (0.684 vs. 0.677), while both methods obtain the same ARI (0.962) and matching score (0.987). Hence, on this elongated dataset, GK-Mapper mainly improves the nontrivial threshold range and the empirical stability radius while preserving essentially the same clustering quality as SFCM.

6.3. Stanford Bunny Dataset

A 3D point cloud of 5000 points sampled from the Stanford Bunny mesh [25] (Open3D), centered, rotated, and ℓ_2 normalized. Both methods use $c = 8$, $t = 0.25$, with $m_0 = 5.000$ (SFCM) and $m_0 = 3.943$ (GK-Mapper). GK-Mapper raises t_{crit} from 0.3762 to 0.4511 and slightly improves r^* from 0.0072 to 0.0096. Under $h = 0.10$, SFCM shows one edge change, whereas GK-Mapper shows no edge change.

The GK-Mapper graph is topologically richer than the SFCM graph. SFCM gives $|V| = 8$, $|E| = 10$, $b_0 = 1$, and $b_1 = 3$, whereas GK-Mapper gives $|V| = 8$, $|E| = 13$, $b_0 = 1$, and $b_1 = 6$; see Fig. 5. The Silhouette

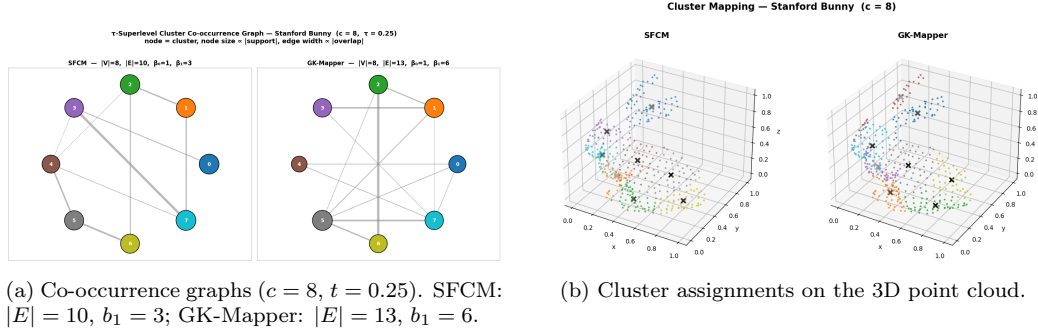


Figure 5: Stanford Bunny dataset.

score is higher for SFCM (0.346 vs. 0.307), while ARI and matching score are not available because the point cloud has no class labels. These results suggest that GK-Mapper retains a more connected and cycle-rich graph on the Bunny point cloud while also reducing the observed edge variation under the selected perturbation.

6.4. UCI Handwritten Digits Dataset

1797 grayscale 8×8 digit images [30, 31], reduced to 20 dimensions via PCA ($\approx 95\%$ variance) and standardized. Both methods use $c = 10$, $t = 0.12$, with $m_0 = 1.40$ (SFCM) and $m_0 = 2.400$ (GK-Mapper). GK-Mapper markedly raises t_{crit} from 0.1000 to 0.4386. Since $t = 0.12$ lies above the SFCM critical threshold, the SFCM graph becomes edgeless, with $|E| = 0$, $b_0 = 10$, and $b_1 = 0$. In contrast, GK-Mapper retains a nontrivial graph with $|E| = 23$, $b_0 = 2$, and $b_1 = 15$.

The empirical stability radius is slightly larger for SFCM (0.0004 vs. 0.0003), and SFCM shows no edge changes under $h = 0.10$, while GK-Mapper gives $|E_{\text{chg}}| = 11$. This comparison must be interpreted carefully because the SFCM graph is already edgeless at the selected threshold; therefore, the absence of edge changes does not represent preservation of a meaningful overlap structure. In terms of clustering metrics, GK-Mapper performs better: the Silhouette score improves from -0.054 to 0.036 , the ARI improves from 0.143 to 0.257 , and the matching score improves from 0.234 to 0.439 . Thus, for UCI Digits, GK-Mapper produces a nontrivial graph and better label agreement, although with a slightly smaller empirical stability radius and more observed edge changes.

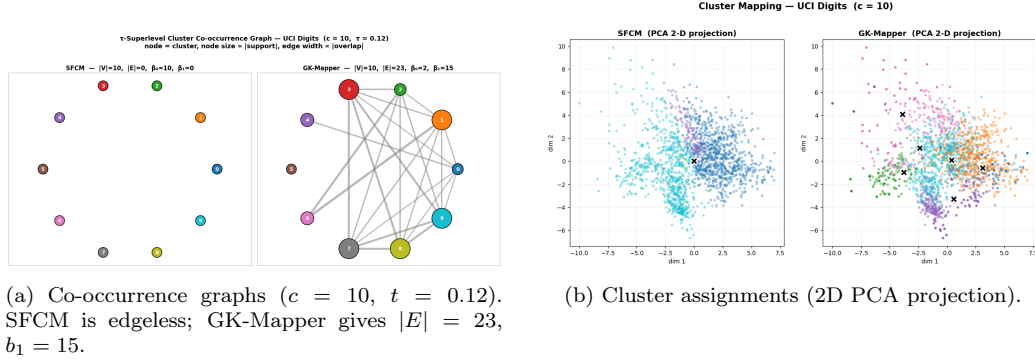


Figure 6: UCI Digits dataset.

6.5. Wisconsin Breast Cancer Dataset

569 samples with 30 features describing cell nuclei from digitised fine-needle aspirates [32, 30, 31]; binary malignant/benign labels. To probe the high-resolution regime ($1/c \approx 0.01$), we use $c = 100$, $t = 0.015$, $h = 0.10$, with $m_0 = 1.65$ (SFCM) and $m_0 = 2.40$ (GK-Mapper). GK-Mapper substantially increases t_{crit} from 0.0291 to 0.2478 and improves the empirical stability radius from 0.0011 to 0.1596. It also sharply reduces the number of edge changes from 879 to 11.

The graph structures are very different. SFCM produces a dense graph with $|E| = 1734$, $b_0 = 2$, and $b_1 = 1636$, whereas GK-Mapper produces a much sparser graph with $|E| = 21$, $b_0 = 94$, and $b_1 = 15$; see Fig. 7. This indicates that the adaptive Gustafson Kessel metric strongly reduces excessive overlap in this high-dimensional biomedical dataset. In terms of clustering metrics, GK-Mapper improves the Silhouette score from -0.203 to 0.019 and the matching score from 0.455 to 0.504 , while SFCM obtains a higher ARI (0.256 vs. 0.143). Thus, GK-Mapper gives a more stable and much sparser graph, although SFCM aligns better with the binary labels under ARI.

7. Discussion

The empirical validation across five datasets shows that GK-Mapper consistently increases the critical threshold t_{crit} compared with SFCM. The increase is observed on the Unit Circle dataset (0.4016 to 0.4652), the Anisotropic Ellipsoidal dataset (0.3723 to 0.4390), the Stanford Bunny dataset (0.3762

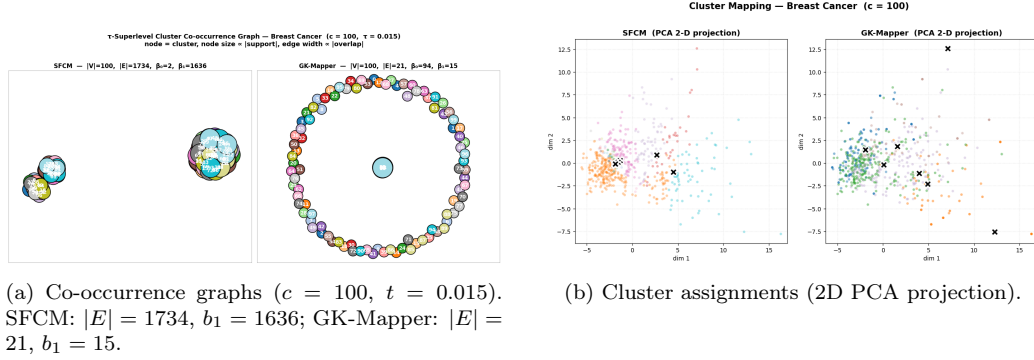


Figure 7: Wisconsin Breast Cancer dataset.

to 0.4511), the UCI Digits dataset (0.1000 to 0.4386), and the Breast Cancer dataset (0.0291 to 0.2478). This indicates that GK Mapper allows the graph to remain nontrivial over a wider range of threshold values. In Theorem 1, a threshold satisfying $t > t_{\text{crit}}$ makes the graph edgless and discrete. Thus, finding t_{crit} is an essential first step before interpreting the resulting Mapper graph.

GK-Mapper also produces a larger empirical stability radius in four of the five datasets. The improvement is most visible for the Unit Circle dataset (0.0833 vs. 0.0229), the Anisotropic Ellipsoidal dataset (0.0333 vs. 0.0132), the Stanford Bunny dataset (0.0096 vs. 0.0072), and the Breast Cancer dataset (0.1596 vs. 0.0011). The only exception is the UCI Digits dataset, where SFCM has a slightly larger radius (0.0004 vs. 0.0003). This supports the local stability result of Theorem 2: when the membership values remain separated from the threshold t , small changes in the fuzzifier m do not alter the thresholded cover, and hence the graph remains unchanged. The empirical results therefore suggest that GK-Mapper often provides a wider local stability region, especially when the data is heterogeneous.

The edge-change results shows that GK-Mapper reduces the number of edge changes on the Breast Cancer dataset, where $|E_{\text{chg}}|$ drops from 879 for SFCM to 11 for GK-Mapper. It also improves stability on the Stanford Bunny dataset, where the edge change decreases from 1 to 0. On the Anisotropic Ellipsoidal and Unit Circle datasets, both methods show no edge changes under the selected perturbation. However, on the UCI Digits dataset, GK-Mapper has 11 edge changes whereas SFCM has none. This case should be interpreted carefully, because the SFCM graph is already edgeless at the

Table 1: Summary of SFCM and GK-Mapper performance across datasets. r^* : empirical stability radius; $|E_{\text{chg}}|$: edge change for $h=0.10$.

Dataset	Method	t_{crit}	r^*	m_0	$ E_{\text{chg}} $	edges	b_0	b_1	Sil.	ARI	Match.
Breast Cancer	SFCM	0.0291	0.0011	1.65	879	1734	2	1636	-0.203	0.256	0.455
	GK-Mapper	0.2478	0.1596	2.40	11	21	94	15	0.019	0.143	0.504
UCI Digits	SFCM	0.1000	0.0004	1.40	0	0	10	0	-0.054	0.143	0.234
	GK-Mapper	0.4386	0.0003	2.40	11	23	2	15	0.036	0.257	0.439
Anisotropic Ellipsoidal	SFCM	0.3723	0.0132	1.20	0	3	1	1	0.684	0.962	0.987
	GK-Mapper	0.4390	0.0333	1.20	0	3	1	1	0.677	0.962	0.987
Stanford Bunny	SFCM	0.3762	0.0072	5.000	1	10	1	3	0.346	NaN	NaN
	GK-Mapper	0.4511	0.0096	3.943	0	13	1	6	0.307	NaN	NaN
Unit Circle	SFCM	0.4016	0.0229	4.457	0	8	1	1	0.495	0.513	0.713
	GK-Mapper	0.4652	0.0833	2.286	0	8	1	1	0.480	0.814	0.913

chosen threshold, while GK-Mapper retains a nontrivial graph with 23 edges. Therefore, the absence of edge changes for SFCM in this case reflects an already-empty graph rather than a more informative stable structure. This agrees with Theorem 3, which states that edge changes are controlled by threshold crossings of membership values.

The clustering metrics show that graph stability and label agreement are related but distinct objectives. GK-Mapper improves the ARI and matching score on the Unit Circle and UCI Digits datasets. For example, on UCI Digits, the ARI increases from 0.143 to 0.257, and the matching score increases from 0.234 to 0.439. On the Unit Circle dataset, the ARI increases from 0.513 to 0.814, and the matching score increases from 0.713 to 0.913. On the Anisotropic Ellipsoidal dataset, both methods obtain the same ARI and matching score, while SFCM has a slightly higher silhouette value. On the Breast Cancer dataset, GK-Mapper improves the silhouette and matching score, but SFCM obtains a higher ARI. Therefore, a more stable or more structured Mapper graph does not automatically imply stronger agreement with external class labels.

Several limitations remain. The theoretical results assume local regularity of the optimisation path but do not prove global uniqueness or global smoothness of FCM or GK-FCM solutions. The stability guarantee is local in the fuzzifier parameter, and the theory does not establish a universal ordering between the stability radius of GK-Mapper and that of SFCM. The

empirical behaviour also depends on the chosen threshold, number of clusters, initialisation, fuzzifier grid, perturbation size, and scatter-matrix regularisation. Therefore, GK-Mapper should not be viewed as uniformly superior to SFCM; rather, it provides a geometry-adaptive alternative that can be more stable and more informative when the data contain anisotropic or heterogeneous structures.

In summary, the proposed framework separates three issues in fuzzy Mapper construction: whether the graph is nontrivial, whether it remains locally stable under perturbations of the fuzzifier, and whether the resulting graph agrees with external labels. The experiments indicate that GK-Mapper usually increases the nontrivial threshold range and often improves local stability, particularly on geometrically complex datasets. At the same time, the comparison with SFCM remains data-dependent, so the choice between the two methods should be guided by the geometry of the dataset and the goal of the analysis.

8. Conclusion

We introduced the Gustafson Kessel Mapper (GK-Mapper) algorithm, that generalises SFCM [5] by replacing its Euclidean cover with an Ellipsoidal cover from the Gustafson-Kessel FCM algorithm [15, 2].

We then developed a stability framework that (i) identifies the edgeless zone boundary $t_{\text{crit}}(m) = \max_i \max_{j \neq k} \min\{u_{ij}(m), u_{ik}(m)\}$, (ii) establishes local graph stability near fuzzifier values, and (iii) bounds edge changes by membership threshold crossings. We also showed that under a single crossing condition, the critical event set satisfies $|\mathcal{T}| \leq nc$, and when $\mathcal{T}$ is finite the graph freezes beyond $m^{**} = \sup \mathcal{T}$.

To validate these theorems, we did experiments on the Circle, Anisotropic Ellipsoidal, Stanford Bunny, UCI Digits, and Wisconsin Breast Cancer datasets. In all these cases, GK-Mapper has yielded more stable regions, while clustering quality remains dataset-dependent. This confirms that GK-Mapper is more effective in terms of graph stability.

Future work includes (i) characterising single-crossing behaviour along the full GK-FCM path, (ii) extending the construction to fuzzy c -varieties [2, Section 23], (iii) integrating persistent homology to track the filtration induced by varying m [10, 13, 14], and (iv) deriving Lipschitz-type bounds linking membership-space perturbations to topological distances [18, 8].

Author contributions

Annesha Sen and Shivam Singh contributed to the conceptualization of the problem and wrote the main manuscript text. S.P. Tiwari supervised the work and helped in the preparation of manuscript.

Declarations

Competing interests: The authors confirm that they have no competing interests.

References

- [1] S. Ben-David, U. von Luxburg, and D. Pál, “A sober look at clustering stability,” in *Proc. 19th Annu. Conf. Learning Theory (COLT)*, 2006, pp. 5-19.
- [2] J. C. Bezdek, *Pattern Recognition with Fuzzy Objective Function Algorithms*. New York, NY, USA: Plenum Press, 1981.
- [3] J. C. Bezdek, R. Ehrlich, and W. Full, “FCM: The fuzzy C -Mean clustering algorithm,” *Comput. Geosci.*, vol. 10, no. 2-3, pp. 191-203, 1984.
- [4] Q.-T. Bui, B. Vo, H.-A. N. Do, N. Q. V. Hung, and V. Snasel, “F-Mapper: A fuzzy mapper clustering algorithm,” *Knowl.-Based Syst.*, vol. 189, p. 105097, 2020.
- [5] Q.-T. Bui, B. Vo, V. Snasel, W. Pedrycz, T.-P. Hong, N.-T. Nguyen, and M.-Y. Chen, “SFCM: A fuzzy clustering algorithm of extracting the shape information of data,” *IEEE Trans. Fuzzy Syst.*, vol. 29, no. 1, pp. 75-89, Jan. 2021.
- [6] M. Carrière and S. Oudot, “Structure and stability of the 1-dimensional Mapper,” *Found. Comput. Math.*, vol. 18, no. 6, pp. 1333-1396, 2018.
- [7] M. Carrière, B. Michel, and S. Oudot, “Statistical analysis and parameter selection for Mapper,” *J. Mach. Learn. Res.*, vol. 19, no. 1, pp. 1-39, 2018.
- [8] F. Chazal, V. de Silva, M. Glisse, and S. Oudot, *The Structure and Stability of Persistence Modules*. Cham, Switzerland: Springer, 2016.

- [9] F. Chazal and B. Michel, “An introduction to topological data analysis: Fundamental and practical aspects for data scientists,” *Front. Artif. Intell.*, vol. 4, p. 667963, 2021.
- [10] D. Cohen-Steiner, H. Edelsbrunner, and J. Harer, “Stability of persistence diagrams,” *Discrete Comput. Geom.*, vol. 37, no. 1, pp. 103-120, 2007.
- [11] R. De Maesschalck, D. Jouan-Rimbaud, and D. L. Massart, “The Mahalanobis distance,” *Chemom. Intell. Lab. Syst.*, vol. 50, no. 1, pp. 1-18, 2000.
- [12] T. K. Dey, F. Mémoli, and Y. Wang, “Multiscale Mapper: Topological summarisation via codomain covers,” in *Proc. ACM-SIAM Symp. Discrete Algorithms (SODA)*, 2016, pp. 997-1013.
- [13] H. Edelsbrunner, D. Letscher, and A. Zomorodian, “Topological persistence and simplification,” in *Proc. IEEE Symp. Found. Comput. Sci.*, 2000, pp. 454-463.
- [14] R. Ghrist, “Barcodes: The persistent topology of data,” *Bull. Amer. Math. Soc.*, vol. 45, no. 1, pp. 61-75, 2008.
- [15] D. E. Gustafson and W. C. Kessel, “Fuzzy clustering with a fuzzy covariance matrix,” in *Proc. IEEE Conf. Decision Control*, 1979, pp. 761-766.
- [16] D. Haşegan *et al.*, “Deconstructing the Mapper algorithm to extract richer topological and temporal features from functional neuroimaging data,” *Netw. Neurosci.*, vol. 8, no. 4, pp. 1355-1382, 2024.
- [17] R. Krishnapuram and J. M. Keller, “A possibilistic approach to clustering,” *IEEE Trans. Fuzzy Syst.*, vol. 1, no. 2, pp. 98-110, May 1993.
- [18] M. Lesnick, “The theory of the interleaving distance on multidimensional persistence modules,” *Found. Comput. Math.*, vol. 15, no. 3, pp. 613-650, 2015.
- [19] L. Li *et al.*, “Identification of type 2 diabetes subgroups through topological analysis of patient similarity,” *Sci. Transl. Med.*, vol. 7, no. 311, p. 311ra174, 2015.

- [20] V. N. Madukpe *et al.*, “A comprehensive review of the Mapper algorithm and its applications across various fields (2007–2025),” *Int. J. Data Sci. Anal.*, vol. 21, 2025.
- [21] M. Nicolau, A. J. Levine, and G. Carlsson, “Topology-based data analysis identifies a subgroup of breast cancers with a unique mutational profile and excellent survival,” *Proc. Natl. Acad. Sci. USA*, vol. 108, no. 17, pp. 7265–7270, 2011.
- [22] O. Rafique and A. H. Mir, “A topological approach for cancer subtyping from gene expression data,” *J. Biomed. Inf.*, vol. 102, p. 103357, 2020.
- [23] M. Riani, A. Cerioli, D. Perrotta *et al.*, “Simulating mixtures of multivariate data with fixed cluster overlap in FSDA library,” *Adv. Data Anal. Classif.*, vol. 9, pp. 461–481, 2015.
- [24] G. Singh, F. Mémoli, and G. Carlsson, “Topological methods for the analysis of high-dimensional data sets and 3D object recognition,” in *Eurographics Symp. Point-Based Graphics*, 2007, pp. 91–100.
- [25] H. J. van Veen, N. Saul, D. Eargle, and S. Mangham, “Kepler Mapper: A flexible Python implementation of the Mapper algorithm,” *J. Open Source Softw.*, vol. 4, no. 42, p. 1315, 2019.
- [26] E. P. Xing, A. Y. Ng, M. I. Jordan, and S. Russell, “Distance metric learning with application to clustering with side-information,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 15, 2002.
- [27] D. Q. Zhang and S. C. Chen, “A novel kernelized fuzzy C -Mean algorithm with application in medical image segmentation,” *Artif. Intell. Med.*, vol. 32, no. 1, pp. 37–50, 2004.
- [28] Y. Zhou *et al.*, “Mapper Interactive: A scalable, extendable, and interactive toolbox for the visual exploration of high-dimensional data,” in *Proc. IEEE Pacific Vis. Symp. (PacificVis)*, 2021, pp. 101–110.
- [29] H. J. Zimmermann, *Fuzzy Set Theory-and Its Applications*, 4th ed. Dordrecht, Netherlands: Springer, 2001.
- [30] D. Dua and C. Graff, “UCI Machine Learning Repository,” University of California, Irvine, School of Information and Computer Sciences, 2019.

- [31] F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *J. Mach. Learn. Res.*, vol. 12, pp. 2825-2830, 2011.
- [32] W. H. Wolberg, W. N. Street, and O. L. Mangasarian, “Breast cancer Wisconsin (diagnostic) data set,” UCI Machine Learning Repository, 1995.

Algorithm 1 Gustafson-Kessel Fuzzy C -Mean (GK-FCM)

Require: Dataset $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^p$, number of clusters c , fuzzifier $m > 1$, volume parameters $q_j > 0$, tolerance tol

Ensure: Membership matrix $U = [u_{ij}]$, cluster centres $V = \{v_1, \dots, v_c\}$

1: Initialise $U^{(0)}$ with $\sum_{j=1}^c u_{ij} = 1$ for all i ; set $t \leftarrow 0$

2: **repeat**

3: **for** $j = 1$ to c **do**

4:
$$v_j^{(t)} = \frac{\sum_{i=1}^n (u_{ij}^{(t)})^m x_i}{\sum_{i=1}^n (u_{ij}^{(t)})^m}$$

5: **end for**

6: **for** $j = 1$ to c **do**

7:
$$S_{f_j}^{(t)} = \sum_{i=1}^n (u_{ij}^{(t)})^m (x_i - v_j^{(t)})(x_i - v_j^{(t)})^\top$$

8:
$$A_j^{(t)} = q_j \cdot \det(S_{f_j}^{(t)})^{1/p} \cdot (S_{f_j}^{(t)})^{-1}$$

9: **end for**

10: **for** $i = 1$ to n ; $j = 1$ to c **do**

11:
$$d_{ij}^{(t)} = \sqrt{(x_i - v_j^{(t)})^\top A_j^{(t)} (x_i - v_j^{(t)})}$$

12: **end for**

13: **for** $i = 1$ to n ; $j = 1$ to c **do**

14:
$$u_{ij}^{(t+1)} = \left[\sum_{k=1}^c \left(\frac{d_{ij}^{(t)}}{d_{ik}^{(t)}} \right)^{2/(m-1)} \right]^{-1}$$

15: **end for**

16: $t \leftarrow t + 1$

17: **until** $\|U^{(t)} - U^{(t-1)}\| < \text{tol}$

18: **return** U, V

Algorithm 2 Shape Fuzzy C -Mean (SFCM)

Require: $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^p$, number of clusters c , fuzzifier $m > 1$, overlap threshold t , tolerance $\text{tol} \in (0, 1)$, maximum iterations $k_{\max}$

Ensure: SFCM graph $G_t = (V, E_t)$

```
1: Initialise  $U^{(0)} = [u_{ij}^{(0)}]$  such that  $u_{ij}^{(0)} \in [0, 1]$  and  $\sum_{j=1}^c u_{ij}^{(0)} = 1$  for all  $i$ 
2: Set  $k \leftarrow 0$ 
3: repeat
4:   for  $j = 1$  to  $c$  do
5:     
$$v_j \leftarrow \frac{\sum_{i=1}^n (u_{ij}^{(k)})^m x_i}{\sum_{i=1}^n (u_{ij}^{(k)})^m}$$

6:   end for
7:   for  $i = 1$  to  $n$ ;  $j = 1$  to  $c$  do
8:     
$$u_{ij}^{(k+1)} \leftarrow \left[ \sum_{\ell=1}^c \left( \frac{\|x_i - v_j\|}{\|x_i - v_\ell\|} \right)^{2/(m-1)} \right]^{-1}$$

9:   end for
10:   $k \leftarrow k + 1$ 
11: until  $\max_{i,j} |u_{ij}^{(k)} - u_{ij}^{(k-1)}| < \text{tol}$  or  $k = k_{\max}$ 
12: Compute  $T_0 \leftarrow \min_{i,j} u_{ij}$  and  $T_1 \leftarrow \min_i \max_j u_{ij}$ 
13: Clamp  $t \leftarrow \max(T_0, \min(t, T_1))$ 
14: for  $j = 1$  to  $c$  do
15:    $C_j(t) \leftarrow \{x_i \in X : u_{ij} \geq t\}$ 
16: end for
17:  $V \leftarrow \{1, \dots, c\}$ ,  $E_t \leftarrow \emptyset$ 
18: for  $j = 1$  to  $c - 1$ ;  $k = j+1$  to  $c$  do
19:   if  $C_j(t) \cap C_k(t) \neq \emptyset$  then
20:      $E_t \leftarrow E_t \cup \{(j, k)\}$ 
21:   end if
22: end for
23: return  $G_t = (V, E_t)$ 
```

Algorithm 3 Gustafson Kessel Mapper (GK-Mapper)

Require: $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^p$, number of clusters c , fuzzifier $m > 1$, threshold t , tolerance tol

Ensure: GK-Mapper graph $G_t(m) = (V, E)$

```
1: Initialise  $U^{(0)}$  with  $\sum_{j=1}^c u_{ij}^{(0)} = 1$  for all  $i$ 
2: repeat
3:   for  $j = 1$  to  $c$  do
4:      $v_j = \frac{\sum_{i=1}^n (u_{ij})^m x_i}{\sum_{i=1}^n (u_{ij})^m}$ 
5:      $S_{f_j} = \sum_{i=1}^n (u_{ij})^m (x_i - v_j)(x_i - v_j)^\top$ 
6:      $A_j = q_j \det(S_{f_j})^{1/p} S_{f_j}^{-1}$ 
7:   end for
8:   for  $i = 1$  to  $n$ ;  $j = 1$  to  $c$  do
9:      $d_{ij}^{\text{GK}} = \sqrt{(x_i - v_j)^\top A_j (x_i - v_j)}$ 
10:  end for
11:  for  $i = 1$  to  $n$ ;  $j = 1$  to  $c$  do
12:     $u_{ij} = \left[ \sum_{k=1}^c \left( \frac{d_{ij}^{\text{GK}}}{d_{ik}^{\text{GK}}} \right)^{2/(m-1)} \right]^{-1}$ 
13:  end for
14: until  $\|U^{(t+1)} - U^{(t)}\|_F < \text{tol}$ 
15: for  $j = 1$  to  $c$  do
16:    $C_j(t) \leftarrow \{x_i \in X : u_{ij} \geq t\}$ 
17: end for
18:  $V \leftarrow \{1, \dots, c\}$ ,  $E \leftarrow \emptyset$ 
19: for  $j = 1$  to  $c - 1$ ;  $k = j+1$  to  $c$  do
20:   if  $C_j(t) \cap C_k(t) \neq \emptyset$  then  $E \leftarrow E \cup \{(j, k)\}$ 
21:   end if
22: end for
23: return  $G_t(m) = (V, E)$ 
```
